

Homework 1

Please submit your homework as a python notebook file. Your file should be named *last_name_firstnameinitial_hw1*. For example, my file would be named *triantafillou_s_hw1.ipynb*.

1. (30 points) Myopia, or nearsightedness, results from a number of genetic and environmental factors. In 1999, Quinn et al studied the relation between myopia and ambient lighting at night (for example, from nightlights or room lights) during childhood.
 - (a) The data were gathered by the following procedure, reported in the study. “Between January and June 1998, parents of children aged 2-16 years ... that were seen as outpatients in a university pediatric ophthalmology clinic completed a questionnaire on the child’s light exposure both at present and before the age of 2 years.” Was this study observational, or was it a controlled experiment?
 - (b) The study found that of the children who slept with a room light on before the age of 2, 55% were myopic. Of the children who slept with a night light on before the age of 2, 34% were myopic. Of the children who slept in the dark before the age of 2, 10% were myopic. The study concluded that, “The prevalence of myopia . . . during childhood was strongly associated with ambient light exposure during sleep at night in the first two years after birth.” Do the data support this statement? You may interpret “strongly” in any reasonable qualitative way.
 - (c) On May 13, 1999, CNN reported the results of this study under the headline, “Night light may lead to nearsightedness.” Does the conclusion of the study claim that night light causes nearsightedness?
2. (70 points) This exercise relates to the PISA data set, which can be found in the file *pisa_scores.csv* on the following link. The dataset has 634 entries with the following columns:
 - index_code*: Index code.
 - expenditure_on_education_pct_gdp*: Expenditure on education as a percentage of GDP.
 - mortality_rate_infant*: Infant mortality rate.
 - gini_index*: Gini index.
 - gdp_per_capita_ppp*: GDP per capita in terms of purchasing power parity.
 - inflation_consumer_prices*: Consumer price inflation.
 - intentional_homicides*: Intentional homicides.
 - unemployment*: Unemployment rate.
 - gross_fixed_capital_formation*: Gross fixed capital formation as a percentage of GDP.
 - population_density*: Population density.
 - suicide_mortality_rate*: Suicide mortality rate.
 - tax_revenue*: Tax revenue.
 - taxes_on_income_profits_capital*: Taxes on income, profits, and capital gains.
 - alcohol_consumption_per_capita*: Total alcohol consumption per capita.
 - government_health_expenditure_pct_gdp*: Government health expenditure as a percentage of GDP.
 - urban_population_pct_total*: Urban population percentage of the total population.
 - country*: Country.
 - time*: Year.
 - sex*: Sex.
 - rating*: Value of PISA Results.

Suppose we are interested in predicting PISA scores. The rest of the variables are predictors.

- (a) Which of the predictors are quantitative, and which are qualitative?
- (b) Do PISA scores differ by sex? Create a visualization that best supports your answer.
- (c) We will now focus on a single year (2006). Create a new data set, called *pisa_scores.2006*, which only includes measurements taken on 2006. Remove the sex variable, and keep only total PISA scores. Which of the variables might be useful in predicting the PISA scores, and why?
- (d) Fit a simple linear regression model with one quantitative predictor of your choice (the one you believe will be the most informative) to predict the PISA scores. Report the coefficients, R-squared value, and the significance of the predictor.

- (e) Fit a multiple linear regression model that includes at least three and at most five predictors. Report the coefficients, R-squared value, adjusted R-squared value, and the p-values for each predictor.
- (f) Visualize the residuals of your final model. Create a residual plot to check for any patterns or deviations from the assumptions of linear regression. Discuss any issues that might affect the model's validity.
- (g) According to your initial analysis, which factors do you think may cause low PISA scores, and why? (write at most two paragraphs)